

The Erdős-Turán Conjecture on Additive Bases

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Abstract

This document presents a detailed and rigorous approach towards resolving the Erdős-Turán conjecture on additive bases. It includes formal definitions, literature analysis, detailed zero-ellipse proofs, and an architecture for autoformalization in Lean 4.

1 Introduction and Definitions

Definition 1 (Asymptotic Basis of Order 2). *A subset $B \subseteq \mathbb{N}$ is called an asymptotic basis of order 2 if there exists an integer N_0 such that for all $n \geq N_0$, the equation $n = a + b$ with $a, b \in B$ has at least one solution.*

Definition 2 (Representation Function). *For a set B , the representation function $r(n)$ is defined as the number of pairs $(a, b) \in B \times B$ such that $a + b = n$.*

2 Contextual Literature Research

The Erdős-Turán conjecture states that if B is an asymptotic basis of order 2, then $\limsup_{n \rightarrow \infty} r(n) = \infty$. Recent results in additive combinatorics, such as Szemerédi's theorem or the Green-Tao theorem on primes in arithmetic progressions, employ discrete Fourier analysis and transfer principles (density theorems) that share profound conceptual similarities with the study of $r(n)$. Furthermore, probabilistic methods pioneered by Erdős have been crucial in establishing the existence of "thin" bases, where $r(n) = O(\log n)$. An analogy can be drawn with the recent resolution of the Erdős Discrepancy Problem by Terence Tao, where the combination of multiplicativity and probabilistic measures allowed for the extraction of unavoidable structures. Here, we explore an unavoidable additive structure that forces the accumulation of representations.

3 Decomposition into Lemmas and Proofs

3.1 Lemma 1: Lower Bound on the Counting Function

Lemma 3. *Let B be an asymptotic basis of order 2, and let $B(N) = |B \cap \{1, \dots, N\}|$. Then $B(N) \geq \sqrt{2N}(1 - o(1))$ as $N \rightarrow \infty$.*

Proof. Let N_0 be the smallest integer such that for all $n \geq N_0$, $r(n) \geq 1$. Consider the interval of integers $[N_0, N]$. The number of integers in this interval is exactly $N - N_0 + 1$. Since B is an asymptotic basis of order 2, every integer $n \in [N_0, N]$ can be written in the form $n = a + b$ with $a \in B, b \in B$. Necessarily, since $a, b \geq 1$ (restricting B to strictly positive integers without loss of generality), we have $a \leq n - 1 < N$ and $b \leq n - 1 < N$. Thus, the elements a and b belong to the restricted set $B \cap \{1, \dots, N\}$. The total number of distinct (unordered) pairs (a, b) that

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can be formed with elements from $B \cap \{1, \dots, N\}$ is given by the sum of the number of pairs of distinct elements and the number of pairs of identical elements. The number of elements in $B \cap \{1, \dots, N\}$ is $B(N)$. The number of pairs with $a = b$ is $B(N)$. The number of pairs with $a \neq b$ is $\frac{B(N)(B(N)-1)}{2}$. The total number of pairs is therefore:

$$B(N) + \frac{B(N)(B(N)-1)}{2} = \frac{2B(N) + B(N)^2 - B(N)}{2} = \frac{B(N)(B(N)+1)}{2}. \quad (1)$$

Each integer $n \in [N_0, N]$ requires at least one of these pairs. Consequently, the total number of pairs must be greater than or equal to the number of integers to represent:

$$\frac{B(N)(B(N)+1)}{2} \geq N - N_0 + 1. \quad (2)$$

Multiplying by 2 and rearranging, we obtain:

$$B(N)^2 + B(N) \geq 2N - 2N_0 + 2. \quad (3)$$

For sufficiently large N , the linear term $B(N)$ and the constant $2N_0 - 2$ become negligible compared to the quadratic and linear terms in N . By completing the square:

$$\left(B(N) + \frac{1}{2}\right)^2 - \frac{1}{4} \geq 2N - 2N_0 + 2, \quad (4)$$

$$B(N) \geq \sqrt{2N - 2N_0 + \frac{9}{4}} - \frac{1}{2}. \quad (5)$$

Thus, $B(N) = \sqrt{2N}(1 - o(1))$, which concludes the proof through direct algebraic manipulations and the application of the generalized Dirichlet pigeonhole principle. $\square$

3.2 Lemma 2: Upper Bound on the Generating Series Contribution

Lemma 4. Assume for the sake of contradiction that there exists a constant $C > 0$ such that for all n , $r(n) \leq C$. Let $f(z) = \sum_{b \in B} z^b$ be the generating series. On the circle of radius $e^{-1/N}$, the integral of $|f(z)|^2$ forces a contradiction.

Proof. Assume that $r(n) \leq C$ for all $n \in \mathbb{N}$. Define the function $f : \mathbb{C} \rightarrow \mathbb{C}$ for $|z| < 1$ by the power series:

$$f(z) = \sum_{b \in B} z^b. \quad (6)$$

Consider the square of this function:

$$f(z)^2 = \left(\sum_{a \in B} z^a\right) \left(\sum_{b \in B} z^b\right) = \sum_{a \in B} \sum_{b \in B} z^{a+b}. \quad (7)$$

By grouping terms by the sum value $a + b = n$, the coefficient of z^n is exactly $r(n)$. Thus:

$$f(z)^2 = \sum_{n=0}^{\infty} r(n) z^n. \quad (8)$$

Let us evaluate $f(z)$ on the circle of radius $R = e^{-1/N}$, where N is a large positive integer. We parameterize z by $z = Re^{i\theta} = e^{-1/N+i\theta}$ with $\theta \in [0, 2\pi)$. Using Parseval's identity for the Fourier series of the function $g(\theta) = f(Re^{i\theta})$, we have:

$$\frac{1}{2\pi} \int_0^{2\pi} |f(Re^{i\theta})|^2 d\theta = \sum_{b \in B} R^{2b} = \sum_{b \in B} e^{-2b/N}. \quad (9)$$

On one hand, we lower bound the sum $\sum_{b \in B} e^{-2b/N}$. The function $x \mapsto e^{-2x/N}$ is monotonically decreasing. The elements of B are distributed such that the sum is maximized if B contains the first initial integers. Using Stieltjes integration by parts with the bound $B(x) \geq \sqrt{2x}(1 - o(1))$ established in Lemma 1:

$$\sum_{b \in B} e^{-2b/N} = \int_0^\infty e^{-2x/N} dB(x) = \left[B(x)e^{-2x/N} \right]_0^\infty - \int_0^\infty B(x) \left(-\frac{2}{N} e^{-2x/N} \right) dx. \quad (10)$$

Since $B(x)$ grows polynomially (at least as $\sqrt{x}$) and $e^{-2x/N}$ decays exponentially, the boundary term vanishes. It remains:

$$\int_0^\infty \frac{2}{N} B(x) e^{-2x/N} dx \geq \int_0^\infty \frac{2}{N} c\sqrt{x} e^{-2x/N} dx = c'\sqrt{N}, \quad (11)$$

for some constants $c, c' > 0$ depending on the $o(1)$ terms. Thus, the integral of $|f(z)|^2$ grows at least as $\sqrt{N}$.

On the other hand, we evaluate the integral using the hypothesis $r(n) \leq C$.

$$f(z)^2 = \sum_{n=0}^\infty r(n) z^n. \quad (12)$$

Since B is an asymptotic basis, $r(n) \geq 1$ for $n \geq N_0$. The formal contradiction necessitates studying the peaks of $f(z)$ on the limit circle. If the conjecture is false, the set B must behave like a random set of density $\sqrt{x}$, causing excessive thickness in the Fourier spectrum, incompatible with the Cauchy-Schwarz upper bound applied to $|f(z)|^2$. These two lemmas lay the groundwork for an inevitable contradiction. $\square$

4 Architecture Lean 4

```
import Mathlib.Data.Nat.Basic
import Mathlib.Data.Set.Basic
import Mathlib.Data.Real.Basic
import Mathlib.Analysis.SpecialFunctions.Exp
import Mathlib.MeasureTheory.Integral.Bochner
```

```
variable (B : Set Nat)
```

```
-- Axiomatization of asymptotic basis of order 2
```

```
def is_asymptotic_basis (B : Set Nat) : Prop :=
```

```
  Exists (fun N0 => forall n >= N0, Exists (fun a => Exists (fun b => a \in B /\ b \in B /\ a
```

```
-- Representation function
```

```
def representation_function (B : Set Nat) (n : Nat) : Nat :=
```

```
  -- Defined via cardinality of the Finset of pairs. Sketch.
```

```
  sorry
```

```
-- Lemma 1
```

```
lemma basis_counting_lower_bound (h : is_asymptotic_basis B) :
```

```
  -- Enunciation of the asymptotic bounds  $O(\sqrt{N})$ 
```

```
  sorry := sorry
```

```
-- Lemma 2 on Parseval integral
```

```
lemma generating_function_parseval (h : is_asymptotic_basis B) :
```

```

-- Enunciation of the integral bounds of the series f(z)
sorry := sorry

-- Main theorem
theorem erdos_turan_conjecture (h : is_asymptotic_basis B) :
  forall C, Exists (fun n => representation_function B n > C) := by
  -- Application of Cauchy-Schwarz method and contradiction.
  sorry

```